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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

AI-Powered Smart Computer Laboratory Inventory Management System

SYSTEM ARCHITECTURE, USE CASE & SEQUENCE DIAGRAMS SPECIFICATION

Document Type: System Engineering Blueprint	**Academic Year:** 2025 – 2026
Lab Facility: AI & DS Computer Labs (F106, F107)	**Frameworks:** FastAPI + React 18 + Scikit-Learn
Target Roles: Instructor, HOD, Principal, Technician, Admin	**Security Standard:** Two-Tier RBAC + Anti-Self-Approval

Executive Overview

The **SMVITM AI & DS Smart Computer Laboratory Inventory Management System** is an institutional software solution designed to replace paper stock ledgers and uncoordinated spreadsheets across computer laboratories. It manages 177+ capital assets (Acer Core i5 Workstations, Dell OptiPlex systems, high-speed networking switches, digital ceiling projectors) and 900+ bulk lab peripherals. The platform features native QR barcode tracking, machine-learning-assisted fault triage, annual stock verification ledgering, and a cryptographically enforced **two-tier inter-lab transfer workflow** requiring Department HOD endorsement and Principal executive sanction.

1. System Architecture & Component Topology

The application is engineered using a modern **Three-Tier Decoupled Architecture** (Presentation Tier, Business Logic Tier, and Data Persistence Tier) coupled with an asynchronous background communication subsystem. This structure guarantees high responsiveness, role isolation, and 100% data integrity.

Architecture Components Explained in Simple English:

1. **Presentation Tier (Client Frontend):** Built with React 18 and Vite. It runs smoothly inside any modern browser (Chrome, Edge, Firefox). It renders interactive dashboards, futuristic glowing instructor docks, live camera QR scanning, and sound-enabled celebratory toasts.
2. **Application Tier (FastAPI Backend):** High-performance Python backend. It handles all institutional logic, enforces security permissions (RBAC), prevents conflicts of interest (anti-self-approval), and predicts fault severities with machine learning.
3. **AI Intelligence Subsystem:** A trained Scikit-learn TF-IDF classifier that analyzes plain English problem statements (e.g., 'Workstation display flickering blue tint') and automatically predicts whether it is Hardware, Networking, or OS/Software, assigning priority.
4. **Data Tier (Relational DB & Ledger):** Contains all authentic SMVITM AI & DS laboratory inventory, maintenance logs, and financial asset values with foreign-key referential integrity.
5. **Asynchronous Communication Gateway:** Dispatches instant in-app alerts and official emails directly to the recipient's institutional mailbox (`principal@sode-edu.in`, `instructor@sode-edu.in`).

2. Comprehensive System Use Case Diagram

The Use Case diagram models the system's operational scope by detailing how each user role (Actor) interacts with specific functional features (Use Cases) within the SMVITM Computer Laboratory ecosystem.

Actor	Core Operational Use Cases (Goals)	Institutional Access Scope
Lab Instructor (Chitrakala)	• Scan Workstation QR & view live specs • Register new bench equipment • Log fault ticket with AI diagnosis • Initiate inter-lab equipment transfer • Physical dispatch & receipt confirmation • View incoming institutional webmail	Core Computing Lab (F106), Networks Lab (F107), Edge Computing Lab
Department HOD (Tejaswini H)	• Review departmental transfer requests • Endorse transfer & forward to Principal • Track laboratory inventory valuation • Monitor fault resolution time • Export CSV departmental stock audits	Department-wide (All AI & DS computer laboratories & servers)
Principal (Dr. Nagaraj Bhat)	• Review endorsed high-value transfers • Grant official Executive Approval • Executive inventory overview across labs • Review campus-wide stock verification • Audit compliance & capital write-offs	Institutional Executive (Entire college academic & administrative labs)
Technician (Hardware/Network)	• Receive AI-triaged repair tickets • Update repair progress & order spares • Log diagnostic benchmarks & repair cost • Mark workstation back to 'In Use'	Operational Maintenance (All laboratories & repair workshops)
System Admin (Thrisha)	• User role provisioning & password resets • System health & audit trail inspection • Database migration & seed management • Administrative exception overrides	Root Administrative (Full system configuration & audit ledgers)

Use Case Interaction Breakdown:

• **Equipment Relocation:** Lab Instructor creates request → HOD approves at department stage → Principal sanctions at executive stage → Instructor dispatches → Destination Instructor receives with condition check.
• **Fault Management:** Instructor flags defect → AI categorizes → Technician inspects → Repaired → Asset status reinstated to 'In Use'.
• **Verification:** Annual physical stock audit compares scanned QR tags with registered ledger, detecting missing or relocated assets automatically.

3. Detailed Sequence Diagrams by Module

Sequence diagrams illustrate the exact chronological exchange of messages between actors, frontend components, backend services, and the database for each core functional module.

Module 1: User Authentication & RBAC Authorization Flow

How it works: The client sends email and password over TLS. The backend verifies the bcrypt password hash against the database. If correct, a signed JSON Web Token (JWT) containing user ID and role is returned. The frontend securely caches the token and personalizes the navigation dock according to user permissions.

Module 2: Smart Asset Registration & QR Code Generation

How it works: When an instructor registers a computer, the backend auto-generates a standardized institutional asset tag. The frontend renders an encoded QR code that can be printed and affixed to the PC chassis. Scanning this QR code from any smartphone or webcam instantly pulls up live hardware specifications.

Module 3: AI-Powered Fault Diagnosis & Ticket Logging

How it works: Instead of requiring instructors to guess diagnostic categories, the natural language description is evaluated by a machine-learning pipeline. The system categorizes whether the defect is Hardware (e.g. blown SMPS), Network, or Software, and dispatches an urgent notification to the lab technician.

Module 4: Two-Tier Inter-Lab Transfer Approval Workflow

How it works: Moving equipment between labs requires dual approval. The instructor initiates the request. HOD endorses it. The system automatically notifies the Principal via official email and web portal. The moment the Principal clicks 'Grant Official Sanction', an instant neon toast pops up on the instructor's screen with sound, and an official confirmation email is dispatched.

Module 5: Equipment Maintenance & Repair Resolution

How it works: Technicians track the entire maintenance lifecycle. Starting a repair updates the asset's live status to 'In Repair', preventing other instructors from attempting transfers. Upon completion, replaced spare parts and costs are permanently logged, and the machine returns to 'In Use'.

Module 6 & 7: Stock Verification Audit & Webmail Notification

How it works: During annual stock audits, staff rapidly scan all workstations. The system automatically reconciles physical scans against the digital ledger, highlighting misplaced or unreturned equipment and generating compliance reports for VTU and NBA accreditation committees.

4. Institutional Security & Compliance Guarantees

- **Role-Based Access Control (RBAC):** Each user role has restricted capabilities. Instructors cannot access administrative dashboards; Technicians cannot approve inter-lab transfers.
- **Anti-Self-Approval Protocol:** A user cannot approve their own transfer request. If Chitrakala requests a transfer, only independent reviewers (HOD Tejaswini H or Principal Dr. Nagaraj Bhat) can endorse it.
- **Data Integrity:** All transfers, faults, and inventory modifications are logged with an immutable audit entry containing user ID, timestamp, and IP address.
- **Fail-Safe Notification Engine:** Notifications are stored directly in the database and delivered live via background polling, ensuring alerts are never lost even if SMTP is unavailable.